

Vacuum-lift wildfire airships: a diligence report

Prepared for technical and commercial diligence · 9 August 2026

A report on a concept, not on a company. No hardware exists. The apparatus that produced these numbers is the asset under examination.

0. How to read this document

This is a diligence report on a **concept**, not on a company, a product, or a prototype. Nothing has been built. There is no revenue, no letter of intent, no flight article, and no test rig.

What exists is a simulation and its supporting apparatus: a physics model, 309 tests, a 151-case deterministic golden baseline, a 73-source catalogue with 12 written notes, an audit of all 89 published claims, and a defect list of thirteen items, two of them closed. That apparatus is the asset under examination. **The correct question for a reader is not "do these numbers work" but "is this the kind of work that would find out if they didn't".**

The report is organised as an argument rather than as a defence: what the idea is worth if it works (§§1–3), then what would have to be true for it to work (§4), then the economics, the regulatory path, and the quality of the evidence behind all of it. **§4.1 is the entry that could end the conversation**, and if you read one section, read that one — but read it after §3, because the size of the prize is what decides whether the obstacle is worth attacking.

Every model figure is cited by key and verified automatically against the model (`tools/check_figures.py`; the build fails on a mismatch). Every external figure names its source. Where nothing is known, the section says so rather than estimating.

1. The thesis in one page

Claim. Aerial firefighting is limited by turnaround, not by drop size. A very large airtanker delivers seventy tonnes in seconds and then spends most of an hour not delivering anything. A vehicle that dips from a lake like a helicopter but carries like a tanker would change the *sustained* rate — the one that decides whether a line holds — by orders of magnitude.

Buoyant flight is the only way to get payload without paying for lift continuously, and it is the only way to build a firefighting aircraft that never has to land. Helium is expensive and leaks; vacuum is free and does not, if the shell can be made light enough — which is what discrete-lattice construction may finally allow, and what §4.1 says has not been shown yet.

What the model says. Three classes, from 100 t to 10,000 t of water. The largest is 876 m long, flies a 45.5-minute cycle at 15 km each way, and delivers 13,183 t/h at 4.59 kWh/t.

What the model does not say. That the vehicle can be built. The model takes hull mass as an assumption and computes forward from it; §4.1 is about that assumption and it does not survive contact with the sources.

Where the genuine invention is. One place: the descent anchor (§3). It solves a real problem that emerged mid-project, it is $58\times$ cheaper than the rotor work it replaces, and it is the only part of this concept that is not a scaling exercise on prior art.

Why the scale is the point. Energy per tonne *falls* as the ships get bigger — 12.53 kWh/t on the smallest against 4.59 on the largest — because buoyancy scales with volume and drag with area. The square-cube law works against nearly every other vehicle and for this one. It is the reason to be interested in the largest class rather than to start with the smallest.

What would make this investable. Not a better simulation. A gram-level mass breakdown of a lattice shell at scale, from someone who builds them. Everything else is downstream of that number and the number is currently missing (§8). It is also cheap to obtain relative to everything it gates: two expert reviews, not a programme.

2. What the model shows

Everything below is conditional on §4.1 — read it as "if a shell can be built, then —". What makes it worth reading anyway is that the operational arithmetic is where the concept is *strongest*, and it is the part a reader can check line by line today.

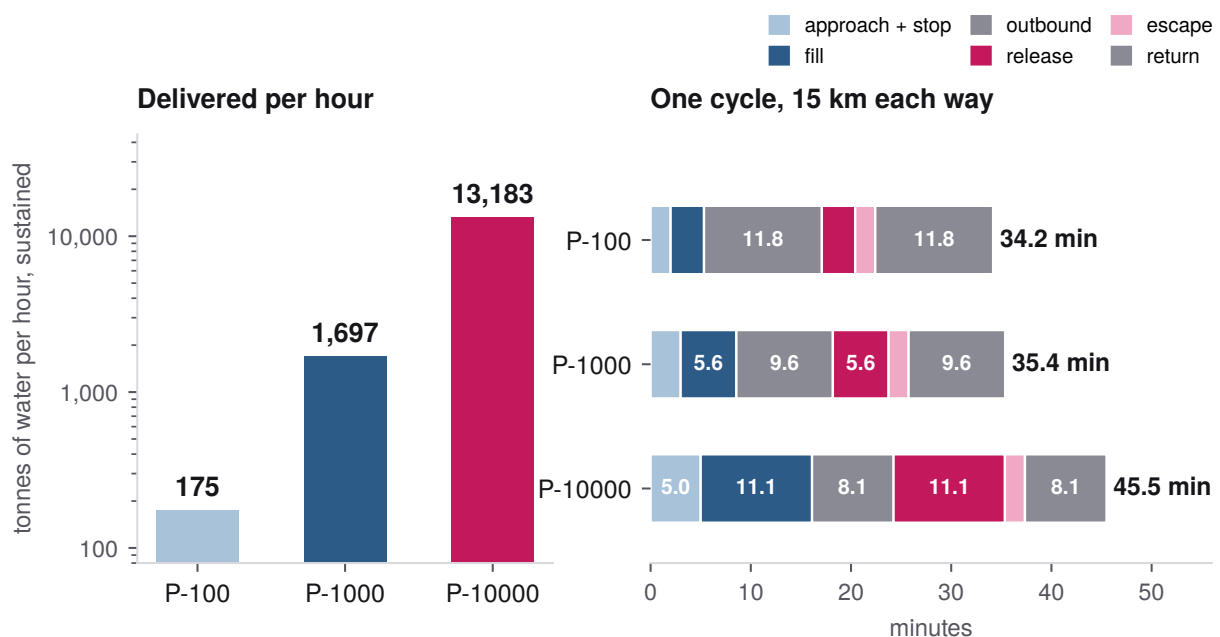


Figure 1. Sustained delivery, and the cycle behind it. The vehicle never lands, so the longest phases are moving water rather than flying.

2.1 The sizing logic is sound and safety-led

Displacement is not chosen to make the numbers work. It is set by a fail-safe requirement:

A hull must be positively buoyant at its working altitude while fully loaded with water and unable to release it.

A ship whose outlets jam rises. For an uncrewed vehicle over a fire, that turns a stuck valve from a crash into an inconvenience. It costs displacement — $2,200\text{ m}^3$ per tonne of payload, giving

+5.25% margin, 21,050.9 t of lift against 20,000 t loaded — and the project pays it rather than trading it away. That is the correct instinct for a vehicle of this size operating autonomously over people.

2.2 The cycle is timed, not asserted

Phase	P-10000, min	What happens
Source approach	5.00	dead stop, lower anchor and pumps
Fill	11.11	15 m ³ /s, no yaw
Outbound	8.14	130 km/h
Release	11.11	one pass
Escape	2.00	buoyancy only, no propulsion
Return	8.14	
Total	**45.51**	10,000 t delivered

Water handling is 22.2 of 45.5 minutes. **The vehicle is a pump with a hull attached**, and at operational ranges the binding constraint is transit distance only because transit is the term that scales with distance. That has a commercial consequence: the value of the concept collapses toward the value of a big pump if the fire is not near water.

2.3 Throughput, if the vehicle exists

	P-100	P-1000	P-10000
Delivered per hour	175 t	1,697 t	13,183 t
Energy per tonne	12.53 kWh	7.40 kWh	4.59 kWh
Energy per cycle	1.253 MWh	7.399 MWh	45.869 MWh

A 747 supertanker drops about 70 t and then flies to a base. Scale is the entire argument, and the square-cube law means only the largest class is interesting: it is $2.3 \times$ cheaper per tonne than the smallest.

3. The one genuine invention, and its exposure

3.1 What it is

The buoyancy that guarantees float-up must be overcome to descend, and it is worst at the bottom, over the water, where the air is densest.

	P-100	P-1000	P-10000
Surplus to hold down at the lake	135.6 t	1,366.9 t	13,722.5 t
Rotor capability	267.2 t	1,318.1 t	12,666.2 t

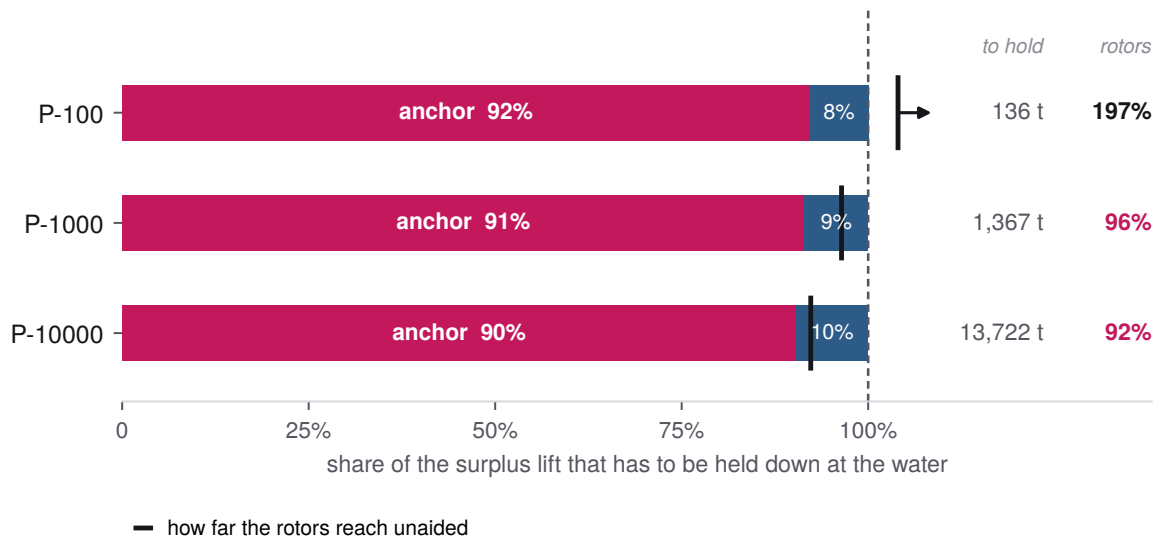


Figure 2. The problem the anchor solves. The bar is what has to be held down at the water; the tick is how far the rotors reach unaided.

The two larger classes **cannot reach their own water source under power**. The solution is to borrow the lake: lower a cable with a collapsible bag, fill it, winch it clear of the surface. 12,400 t of hanging water is 121.6 MN of downward force for the 15 m of lift needed to break the surface — **0.596 MWh** against the 34.5 MWh of rotor work it replaces.

The leverage is in the exponent: induced rotor power goes as thrust^{1.5}, so the letdown term falls from 34.20 MWh to 1.420 MWh — 96%. Alternatives, costed and rejected in the open: nitrogen ballast **475 MWh**, a 1,350 m hose **44 MWh** at 2 m bore and 140 bar, retention **directly reduces the product**.

The mechanism cannot be over-sized: the most water a ship can lift is its own surplus lift, so the physics supplies its own ceiling. And it pays on every class — removing the P-100's bag, which it does not need, costs 34% more energy per cycle.

3.2 Its exposure

Risk Status

Scale 12,400 t against the largest bucket ever built, 9,800 L. **1,265×**. The principle is 43 years old; the engineering

Cable 121.6 MN needs ~440 mm of UHMWPE massing 125 t — **not charged as dry mass anywhere in the mo**

Pend 12,400 t swinging on one cable under an 876 m hull. Not modelled at all.

Rotor The P-10000 has ~79 kg/m² disc loading against a Black Hawk's ~47, across 14 rotors whose combined d

wash
on
wa-
ter

Station The mechanism requires a dead stop, no yaw while lines are down, and departure only when pumps cle
keeping

Positive IP note: this is the one part of the concept that is not obvious from prior art, and it is currently published openly. If there is defensible IP here, it is being given away. That is a deliberate consequence of the open-source strategy and should be a conscious decision rather than a side effect.

4. What would have to be true

Everything in §§2–3 is arithmetic on assumptions. Four of those assumptions were checked against the literature and came back contradicted, and they are stated here in full, with their arithmetic, because a diligence reader should not have to find them.

Two of the four were corrected on 2026-08-09, within hours of being found, because neither had a defensible reading, and both corrections made the model's numbers worse. They are kept here rather than moved to an appendix: what a project did when its own evidence went against it is worth more to a reader than the corrections themselves. **The two that remain open are §4.1 and §4.4. §4.1 is the one that could end this**, and §4.4 changes what the product is rather than whether it can exist.

4.1 The mass budget fails against four independent sources — SEVERE

The model's foundational assumption is $\text{dryT} = \text{payloadT}$: the ship's entire dry mass — shell, skin, joints, rotors, tanks, batteries, pumps, cable — equals the water it carries. For the P-10000 that is 10,000 t inside 22,000,000 m³, a hull-average density of **0.455 kg/m³**.

Source	What it gives	Against our 0.455 kg/m ³
Jenet et al. 2019 (NASA NTRS)	bare discrete-lattice shell, 0.508 kg/m³ , at every radius (their Table 2)	12% over our entire dry allowance
our own cited structural precedent		
Metten & Palazzotto 2013	only real-materials design has structure/buoyancy 0.94	structure alone consumes what
Akhmetiev & Gavrilin 2021	had fraction 0.1 , shell 1.16 kg/m ³	against our implied 0.5
Chin et al. 2021 / Lvovich 2020 (NASA)	X-57 flew 149 Wh/kg at pack level from 225 Wh/kg cells	our 2,000 MWh battery masses

Four sources, four different objections, one conclusion. The 2,000 MWh battery figure deserves emphasis: at the density NASA has actually flown, the battery by itself exceeds the entire dry mass allowance before any structure exists. Even at 500 Wh/kg — the point Lvovich says NASA sees no clear path past — the pack consumes 40% of a budget the hull already exceeds on its own.

And there is a fifth objection that attacks the method rather than the numbers. Both structural sources hold shell mass fraction constant with radius — the scale invariance that lets a metre-scale demonstration imply a 219 m hull. Derveni et al. (2024) show that in pressurised shells with distributed defects, **the single worst defect sets the buckling knockdown factor**, and removing all the lesser defects changes it by under 4%. Defect count is not invariant with radius: it grows with the number of manufactured features, and so does the expected severity of the worst one. **The scale invariance both our structural sources assume cannot hold.**

Diligence status: unresolved, and not resolvable by more modelling. Requires a mass breakdown from a builder of lattice structures. This is the project's single point of failure.

4.2 The solar skin required 76% conversion efficiency — CORRECTED 2026-08-09

The model credited the skin a flat **200 W/m² of electrical output**, continuously, in five separate source files. NRCan's dataset gives 6.34 kWh/m²/day mean July horizontal insolation across eight BC interior fire-belt towns: **264 W/m² incident**, day-averaged. 200 out of 264 is 76% conversion — three and a half times the best cell ever made in a laboratory.

It is ****45 W/m²**** now: 264 × 0.21 flexible module × 0.81 for curvature, cell temperature, soiling and conversion, on a projected area.

	solar	deficit/cycle	endurance
P-100	1.20 → **0.27 MW**	0.32 → **1.10**	35.4 → **10.4 h**
P-1000	5.60 → **1.26 MW**	3.09 → **6.66**	22.9 → **10.6 h**
P-10000	24.00 → **5.40 MW**	24.81 → **41.77**	61.1 → **36.3 h**

Diligence status: **closed, and the finding underneath it is not.** Two things a reader should take from this. First, the correction cut published endurance by roughly two thirds and the project shipped it the same day — that is the behaviour the rest of this report is asking you to price. Second, **no published throughput or energy-per-tonne figure moved at all**, because generation is not in the ledger that computes them. A 4.4× error in the vehicle's power supply was invisible to every test in the repository. That is §8's defect 6, it is still open, and it is the more important half of this entry.

4.3 The nitrogen recovery was thermodynamically impossible — CORRECTED 2026-08-09

$r_{tLN2} = 0.50$ against 'eLN2 = 0.45' kWh/kg recovered **225 kWh per tonne** of liquid nitrogen. The physical exergy of LN2 at 1 bar against a 288 K ambient is **173.4 kWh/t** (Arnaiz-del-Pozo et al. 2020; independently corroborated at 205–214 kWh/t under more favourable assumptions). The term returned 1.3× the work the liquid contains, before any turbine or generator efficiency.

It is **0.20** now — 90 kWh/t, 52% of the exergy, about what a cryogenic expander returns with no external heat source.** The P-10000's cycle rose **43.019 → 45.869 MWh****, 6.6%. Two tests, one on each copy of the constant, now fail the build if $r_{tLN2} \times e_{LN2} \times 1000$ exceeds 173.4.

Diligence status: **closed.** Small in magnitude, absolute in kind, and worth reading for what it says about how the model was being checked before: a unit test existed for this line and it was asserting the violation as correct behaviour. The test now checks the physical ceiling as well as

the value, because the first is a choice and the second is not.

4.4 The headline metric may be measuring the wrong thing — strategic

Two findings from the wildfire-aviation literature, both aimed at the top of the funnel.

The drop may not arrive. The US Forest Service's 2022 assessment states that a drop released 1,000 ft above ground/vegetation "would completely dissipate". ALT.drop is 450 m — **1,476 ft**. The altitude was raised for hull-clearance reasons that are sound (an 876 m hull cannot manoeuvre out of a surprise over a fire), and **the delivery consequence was never costed**. The model has no droplet physics at all: it moves tonnes from a tank to a coordinate.

Tonnes is not the industry's metric. AFUE observed 27,611 drops at 272 incidents. Overall probability of success is 0.82, but for airtankers used to halt fire spread it falls to **0.55–0.67**, and **without ground engagement it is 0.56 against 0.72 with it** — the modal outcome for a drop with no ground crew being *not effective*. AFUE never counts tonnes.

An uncrewed fleet delivering enormous tonnage without ground engagement is, on the only large field study available, **operating in the least effective mode the study measured**.

Diligence status: neither is a modelling error — the model is honest about being a delivery simulator. Both bear on whether the product is valuable. A fire-agency reader will raise them in the first five minutes, and there is currently no answer.

5. Energy, and the only economics the model supports

5.1 The ledger

P-10000, 15 km each way, printed from the model:

Term	MWh
Return transit (drag + cryogenic plant)	14.705
Pumping	10.900
Hotel + manoeuvring	10.689
Outbound transit	9.459
Lakedown	1.420
Anchor	0.596
Nitrogen recovery (credit)	-1.900
Total	**45.869**

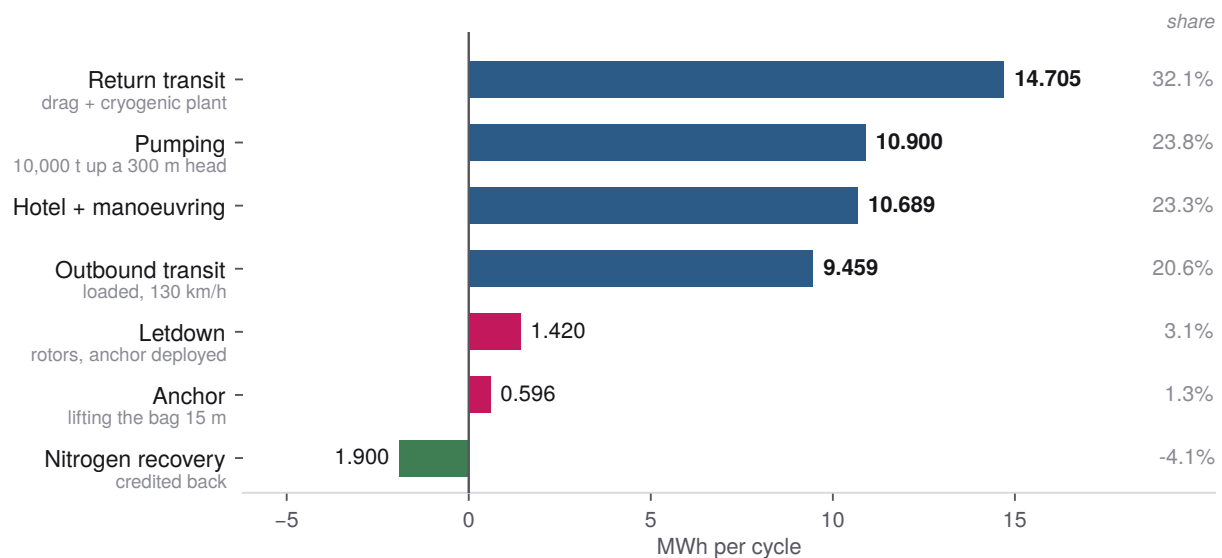


Figure 3. The cycle ledger, printed from the model. Seventy-six per cent of it is transit and pumping; the mechanism that closes the descent costs 1.3%.

5.2 The fleet is a battery being spent

Every class runs a deficit every cycle. This is the project's central public conclusion and it is stated on the site rather than hidden:

	deficit/cycle	endurance
P-100	1.10 MWh	10.4 h
P-1000	6.66 MWh	10.6 h
P-10000	41.77 MWh	36.3 h

Those figures are post-correction and roughly a third of what this report would have carried a day earlier. There is no configuration in which this fleet sustains itself; it requires an energy import chain, and that chain is the business.

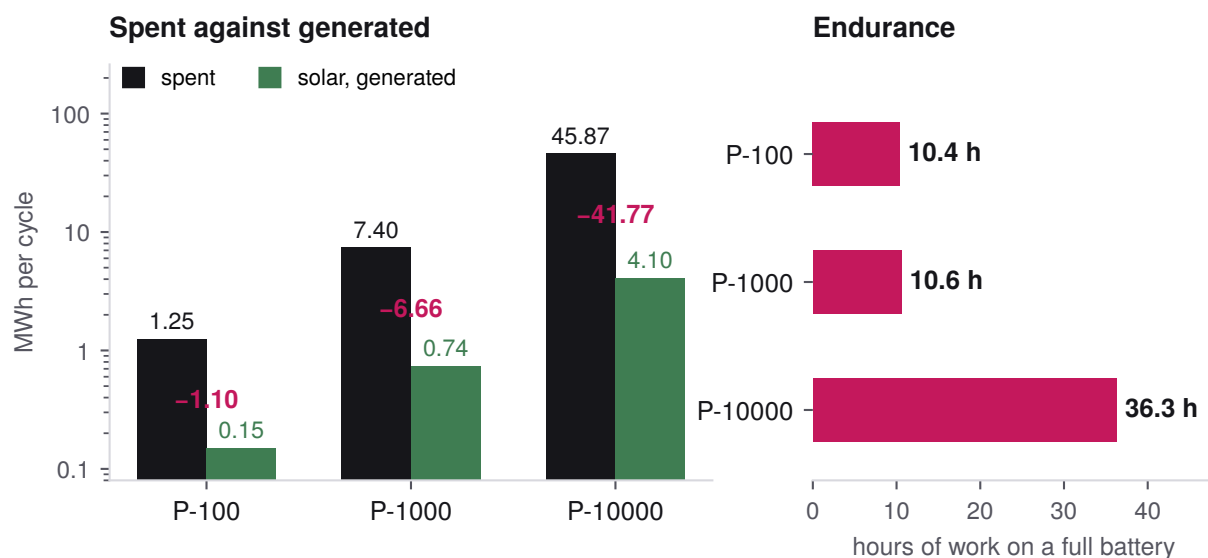


Figure 4. Spend against generation, and endurance on a full battery. This is the project's central public conclusion and it is not a favourable one.

5.3 Grid implications — the number that should govern the conversation

The model publishes no cost, and nothing below is a model output. These are derived from model energy figures and the published BC Hydro Transmission Service Rate Schedule 1830 (effective 1 April 2026: demand \$12.178/kV·A, energy 4.914 ¢/kWh). Assumptions stated inline.

- **Continuous draw per P-10000 in sustained operation:** 45.869 MWh per 45.51-minute cycle = **60.5 MW average**. Net of the corrected solar (5.40 MW), **55.1 MW imported**. That is 9% worse than this report said before the 2026-08-09 corrections, and the earlier figure was the optimistic one.
- **Energy cost per tonne delivered** at 4.914 ¢/kWh: **\$0.23/t** for the P-10000 (4.59 kWh/t), **\$0.62/t** for the P-100 (12.53 kWh/t). Per 10,000 t drop: **~\$2,250**.
- **A ten-ship P-10000 fleet** in continuous operation imports roughly **550 MW**. At Schedule 1830 that is on the order of **US\$26M per month** — about \$20M energy and \$7M demand — assuming unity power factor and continuous operation. Given to one significant digit because the duty-cycle assumption dominates it.

The governing comparison: BC Hydro's total generating capacity is **13.4 GW** (Site C fully operational). A single P-10000's 1,400 MW battery discharge peak is **over 10% of the province's entire generating capacity**. A ten-ship fleet's sustained import is **4.1% of provincial capacity**, continuously, during fire season — which is also peak demand season.

That is not a tariff problem, it is an interconnection and generation-planning problem, and it has a lead time measured in years. BC Hydro's 2025 capacity call drew 106 submissions totalling 19 GW, which indicates the queue this would join.

Not costed anywhere, and material: capital cost of a hull (no basis exists), the megawatt-scale charging infrastructure at each operating base, battery replacement cycles, and the cost of the fire-season duty factor — a fleet sized for August is idle in February.

6. Regulatory and legal

Questioning

Taking the water
The lake's capacity
Flying it
Autonomous as-
sur-
ance
Weather mod-
i-
fi-
ca-
tion
law

IC's *Water Sustainability Act* permits diverting unrecorded water for firefighting **without authorisation**, the

The Act says nothing about whether a lake can stand repeated 10,000 t draws. A P-10000 removes 13,183

CAO Chicago Convention Art. 1 (sovereignty) and Art. 8 (pilotless aircraft need special authorisation o

Any 2024 AI safety-assurance roadmap, EASA AI roadmap 2.0, NIST AI RMF, runtime-assurance literat

ENMOD catalogued. Not obviously engaged by water delivery.

The regulatory path is the second-longest pole after the structure, and unlike the structure it cannot be shortened by a good result in a lab.

7. Quality of the evidence base

This is where the project is genuinely strong, and it is the reason the §4 findings exist at all.

Artefact what it is

sim/The model. No DOM, network, storage, wall clock or location — enforced by a boundary linter across 80
test 206 tests in 39 suites, plus 103 more that need node. **Two fail on purpose**, marked knownFail, so a doc
test 151 class/mode/distance/wind combinations, diffed on every run. Independently reproduced by a sepa
research 73 sources: 3 lead-bearing, **9 that contradict us**, 20 supporting, 41 context. 29 redistributable PDFs with
research 12 notes, each written from the source PDF rather than its abstract, each stating where the source does n
research Audit of all 89 published claims.
map.md
research Every published figure, regenerated from the live model. tools/check_figures.py fails the build if
docs Thirteen defects, each with cost, options and a recommendation. Two closed, eleven open.
QUESTIONS.md

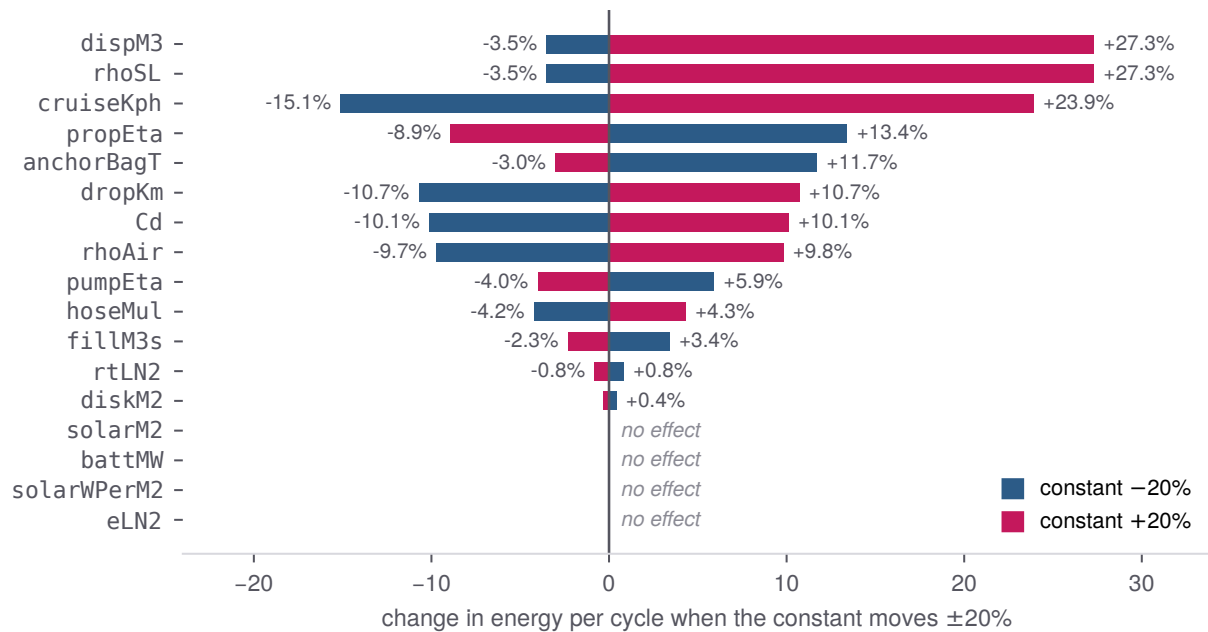


Figure 5. Every constant moved $\pm 20\%$, generated rather than transcribed. Where the model's uncertainty actually lives: aerodynamics and speed, not lift.

The signal to weigh. The defect list went from six to thirteen the day the project audited itself properly — and it is the same model. Nothing was introduced; it was all already true and unnoticed. Six were found by the authors, two by auditing claims against code, four by reading the sources. That ratio is the argument for the process and the warning about the numbers.

The two largest errors ever found in this model were both accounting, not physics. The nitrogen recovery was netted against the pump bill under a $\max(0, \dots)$ and silently deleted the entire pumping cost on two of three classes; and the lift ledger bought air at sea level and spent it at altitude, overstating lift by 22%. Neither was caught by a test. Both were caught by re-deriving a published number from scratch.

Where the process is still weak: 32 of 89 published figures rest on constants with no stated justification, against 15 genuine documented assumptions. Two undocumented inline drag multipliers are worth 16.9 MWh a cycle, while the defect flagged in the README since the first commit is worth 1.4. The project has been auditing the things it knew to doubt.

And the test suite had a hole in it that is worth understanding. 103 of the tests need node, which is not installed on the machine this is developed on, so `make test-node` printed "SKIPPED" and those tests were only ever executed by CI. Two breaks reached the remote that way — including a unit test asserting the impossible nitrogen recovery in §4.3 as correct behaviour. They run in a browser now, against the same files, and `make check` covers them. A reader should weigh both halves: a gap that let a wrong assertion sit unchallenged, and a project that closed it with a tool rather than a resolution.

8. What would have to be proven, in order

1. **A lattice vacuum shell at $\leq 0.455 \text{ kg/m}^3$ including everything.** Nothing in the literature is there. Not resolvable by modelling. Requires a gram-level mass breakdown from someone who builds these structures, plus an answer to the Derveni scale-invariance objection (§4.1). **This is the only §2 finding that both remains open and is capable of ending the concept** — §4.2 and §4.3 are closed, and §4.4 changes what the product is rather than whether it can

exist.

2. **A battery at ≥ 200 Wh/kg pack level, with the rest of the vehicle free.** Not available; NASA has flown 149 and sees no clear path past 500.
3. **A drop from a height an 876 m hull can safely use that still arrives as water.** Currently contradicted by USFS guidance. Needs droplet physics and, eventually, a drop test.
4. **A 12,400 t suspended bag, its cable, its pendulum dynamics, and hull station-keeping over water in wind.** The mechanism is sound in principle and unbuilt at $1,265\times$ the precedent.
5. **An interconnection.** ~ 50 MW continuous per large ship, in a 13.4 GW province, during peak season.
6. **That tonnes delivered is the right product.** AFUE says effectiveness turns on ground engagement, which an uncrewed fleet does not have.

Items 1 and 2 are structural facts about materials that no amount of engineering effort inside this project can change. **They should be resolved before anything else is funded**, and they are cheap to resolve relative to everything downstream: they are literature and expert review, not hardware.

9. Assessment

On the concept. Unproven and currently contradicted at its foundation. The mass premise fails against the project's own cited precedent by 12% before anything is fitted, and the battery alone exceeds the whole dry allowance at demonstrated pack densities. Neither is a detail; both are the first line of the ledger. A reader looking for a reason to stop has one, in §4.1, sourced.

On the invention. The descent anchor is real, is not obvious, solves a problem the project discovered rather than invented, and is $58\times$ cheaper than the alternative. It is worth attention on its own terms — including, possibly, on vehicles other than this one. It is also published openly and therefore unprotected.

On the work. The apparatus is better than the concept. A model that fails its own audit in public, keeps two tests failing on purpose, catalogues nine sources that contradict it, and grows its defect list from six to thirteen by checking properly is doing the thing most concept work avoids. The clearest evidence is what happened on 2026-08-09: two of the four findings in §4 were corrected within hours of being found, both corrections made the published numbers worse — cycle energy up 6.6%, endurance down by roughly two thirds — and both shipped to the live site the same day. If the question is "will this team find out whether the idea works", the evidence is strongly yes. If the question is "does the idea work", the evidence today is **no, on the mass budget**, and the project says so itself in `docs/OPEN-QUESTIONS.md` #11.

Recommended next step, at minimal cost: commission a lattice-structure mass breakdown at 100 m and 219 m radius from a group that builds them, and an independent pack-level mass estimate for a 2,000 MWh flight battery. If either comes back where the literature suggests, the concept needs a different vehicle — smaller, or hybrid-lift, or not buoyant at all — and finding that out costs two reviews rather than a programme.

10. Sources and reproduction

```
" git clone github.com/pinkrobotics/airships && cd airships && make check
"
```

make golden diffs 151 cases; make test runs 206 tests including two deliberate failures and make test-node a further 103; make factsheet regenerates every published figure from the live model; tools/check_figures.py fails the build if this document disagrees with it. ?seed=7&data=snapshot reproduces any run on the live site exactly.

Full catalogue: research/sources.json. Notes: research/notes/. Claim audit: research/evidence-map.md. Defect list: docs/OPEN-QUESTIONS.md. Physics derivations: docs/PHYSICS.md. Public summary of the evidence base: pinkrobotics.ca/research.

Load-bearing sources: Jenett et al. 2019 (NASA NTRS, redistributable — verified by hash against the NTRS original); Akhmeteli & Gavrilin 2021 (CC BY); NOAA/NASA/USAF U.S. Standard Atmosphere 1976.

Sources that contradict us: Metlen & Palazotto 2013; Derveni et al. 2024; Arnaiz-del-Pozo et al. 2020; NRCan 2020; USFS AFUE 2020; USFS 2022; Suter 2005; Chin et al. 2021; Lvovich 2020.

Every figure marked in this document is generated by the model and verified automatically. External figures name their source. Derived commercial figures in §5.3 state their assumptions and are not model outputs — the model publishes no cost.